

ZYRO

LONG-RANGE VIDEO TRANSMISSION

5.8 GHz FIELD VALIDATION REPORT



Document version	V2.0 - Customer Edition
Test date	21 July 2026
Firmware	fw-20260717-4uu8.bin
Prepared for	Customer technical evaluation

Document item	Details
Document title	ZYRO 5.8 GHz Long-Range Video Transmission Field Validation Report
Document ID	ZYRO-TR-5G8-20260721-V2.0
Version	V2.0 - Customer Edition
Test date	21 July 2026
Source basis	Original three-page field test record plus two runtime screenshots supplied with this report
Product designation	5.8 GHz wireless video transmission system; specific commercial model not stated in the source record

Executive summary

MAXIMUM TESTED DISTANCE 9.78 km Open-path field test	DIRECTIONAL ANTENNA 512 / 1024 kbps Smooth at both rates	OMNI ANTENNA 512 kbps Smooth at 9.78 km	OBSERVED LIMITATION 1024 kbps Minor stuttering with omni
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The supplied field record documents successful video transmission at 3.86 km and 9.78 km. At 9.78 km, the directional-antenna configuration was reported smooth at both 512 kbps and 1024 kbps. The omni-antenna configuration was reported smooth at 512 kbps and showed minor stuttering at 1024 kbps.

Data integrity note: This customer edition reorganizes and explains the supplied evidence. It does not replace missing continuous RF logs, packet-loss records, weather data, GPS coordinates or calibrated antenna-gain measurements.

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01

TEST OBJECTIVE AND SCOPE

What was evaluated and what was not recorded

Objective

Evaluate the practical long-range video transmission performance of the supplied 5.8 GHz wireless link at two field distances and compare directional and omni antenna behavior at the longer route.

Evaluation points

- Link establishment and end-to-end video availability at 3.86 km and 9.78 km.
- Observed video smoothness under the tested antenna and bitrate combinations.
- Instantaneous runtime indicators captured from the device web interface.
- Suitability of directional versus omni antennas for the tested long-range use case.

Scope boundaries

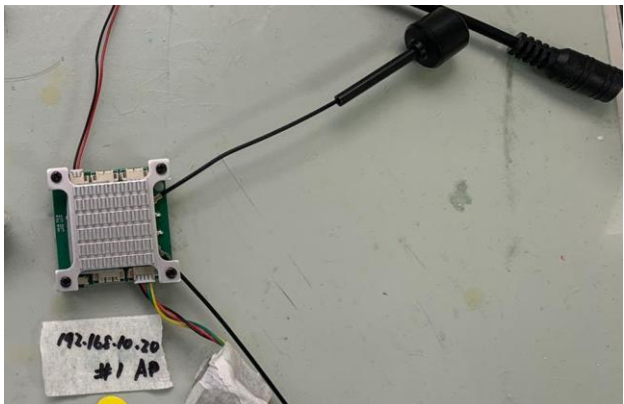
Scope item	Report treatment
Recorded in source	Test date, firmware, main radio settings, endpoint locations, installation height, route distance, antenna type, selected video bitrates and qualitative video result.
Not recorded	Weather, wind, humidity, exact GPS coordinates, antenna gain, cable loss, EIRP verification, continuous RSSI/SNR logs, packet-loss percentage, throughput benchmark and Fresnel-zone survey.
Interpretation rule	Engineering comments in Section 08 are identified as interpretation and are kept separate from the source-derived test results.

Path condition: The photographs and maps show open outdoor paths. A formal line-of-sight and Fresnel-clearance survey was not included in the source package; therefore this report uses the term “open-path field test” rather than claiming verified Fresnel clearance.

02

SYSTEM UNDER TEST

AP, STA, camera and antenna arrangement



AP-side module and antenna connections



STA-side module, camera and omni antennas

Element	Source-supported description
AP side	Wireless module configured as AP. The source photos show both omni and directional antenna options.
STA side	Wireless module configured as STA and connected to a camera. The source photos show dual omni antennas.
Video interface evidence	The supplied runtime screenshots show an ONVIF/ODM interface reporting H.264 at 1920 × 1080.
Antenna comparison	The 9.78 km route compares a directional antenna with an omni antenna under 512 kbps and 1024 kbps video settings.

The specific camera model, antenna gain, cable type and commercial product model are not stated in the source report.

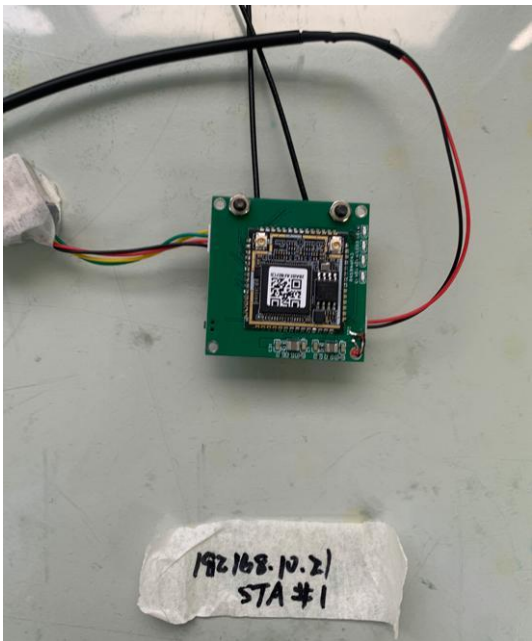
03

TEST CONFIGURATION

Radio and firmware settings recorded in the source



AP radio module



STA radio module

Parameter	Recorded value
Firmware	fw-20260717-4uu8.bin
SSID shown in configuration screen	ZMESH_LINK
Operating band	5.8 GHz
Channel setting	36
Bandwidth	20 MHz
MCS	MCS0
Power setting	31.5 dBm / 3150 mBm - "Use professional antenna" shown in the UI
Baud rate	115200

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CONFIGURATION INTERFACE

Original configuration screen retained as source evidence

SSID:	ZMESH_LINK
Password:	12345678
Frequency:	36
Channel:	5.8G
Baud Rate:	115200
Bandwidth:	20M
MCS:	MCS0
Power:	31.5 dBm / 3150 mBm - Use professional antenna

Figure 3-1. Configuration interface captured in the original source report

Terminology note: The source interface labels the value “36” under Frequency and “5.8G” under Channel. For technical readability, this report presents them as channel 36 and the 5.8 GHz operating band.

04

TEST ROUTES AND ENVIRONMENT

Field locations documented in the source report



Route 1: Shuiyue Bridge to Tiande Temple - 3.86 km



Route 2: Nanliao Fishing Port to Potou Fishing Port - 9.78 km

Route item	Recorded information
Route 1	AP at Shuiyue Bridge (水月橋); STA at Tiande Temple (天德堂); distance 3.86 km.
Route 2	AP at Nanliao Fishing Port (南寮漁港); STA at Potou Fishing Port (坡頭漁港); distance 9.78 km.
Terrain evidence	Route 1 combines a low-elevation bridge site with a higher temple site. Route 2 follows an open coastal path.
Environmental records	Weather, wind, humidity and spectrum occupancy were not recorded in the supplied report.

05

TEST 1 - 3.86 KM

AP at Shuiyue Bridge; STA at Tiande Temple



AP installation at Shuiyue Bridge



STA installation at Tiande Temple

DISTANCE	AP INSTALLATION	STA INSTALLATION	OBSERVED RESULT
3.86 km	~1.5 m	~1.5 m	Smooth
Map measurement in source	Site elevation ~8 m	Site elevation ~85 m	Video reported fluent

Test item	Recorded value
AP antenna	Directional antenna
STA antenna	Omni antenna
Video bitrate	Not stated in the source result section
Video result	影像流暢 / smooth video
Referenced evidence video	Video_260721155939T TX 天德堂&RX 水月橋上.mp4

Result statement: The source report records smooth video for this 3.86 km setup. No continuous latency, RSSI, SNR or packet-loss statistics were included for this route.

06

TEST 2 - 9.78 KM

Directional versus omni antenna comparison



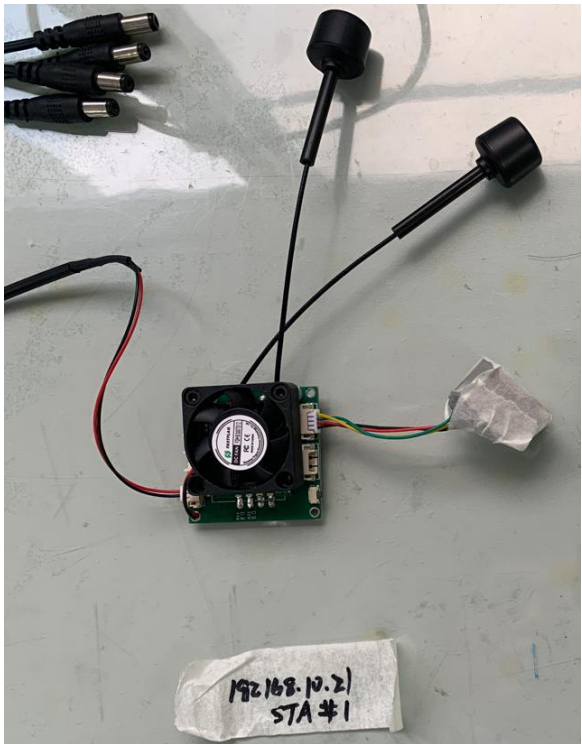
AP installation at Nanliao Fishing Port



STA installation at Potou Fishing Port



Directional antenna used for the long-range comparison



Omni antenna arrangement shown in the source setup

DISTANCE	DIRECTIONAL / 512	DIRECTIONAL / 1024	OMNI / 1024
9.78 km	Smooth	Smooth	Minor stutter
Coastal open-path route	Source-reported result	Source-reported result	Source-reported result

06A

RESULT MATRIX

All test outcomes consolidated from the source record

Test route	Distance	Antenna configuration	Video bitrate	Observed result
Shuiyue Bridge → Tiande Temple	3.86 km	AP directional / STA omni	Not stated in source report	Smooth video
Nanliao Fishing Port → Potou Fishing Port	9.78 km	Directional antenna	512 kbps	Smooth video
Nanliao Fishing Port → Potou Fishing Port	9.78 km	Directional antenna	1024 kbps	Smooth video
Nanliao Fishing Port → Potou Fishing Port	9.78 km	Omni antenna	512 kbps	Smooth video
Nanliao Fishing Port → Potou Fishing Port	9.78 km	Omni antenna	1024 kbps	Minor stuttering

Long-route comparison

Comparison item	Result
Directional antenna	Both 512 kbps and 1024 kbps were reported smooth at 9.78 km.
Omni antenna	512 kbps was reported smooth; 1024 kbps showed minor stuttering.
Referenced directional video	Video_260721162958 指向天線 TX 坡頭&RX 南寮.mp4
Referenced omni video	Video_260721164457 圓頭 ANT TX 坡頭&RX 南寮.mp4

No quantitative acceptance threshold was supplied. Accordingly, this report presents the observations without assigning a formal pass/fail certification.

07

RUNTIME EVIDENCE

Instantaneous device status and ping captures

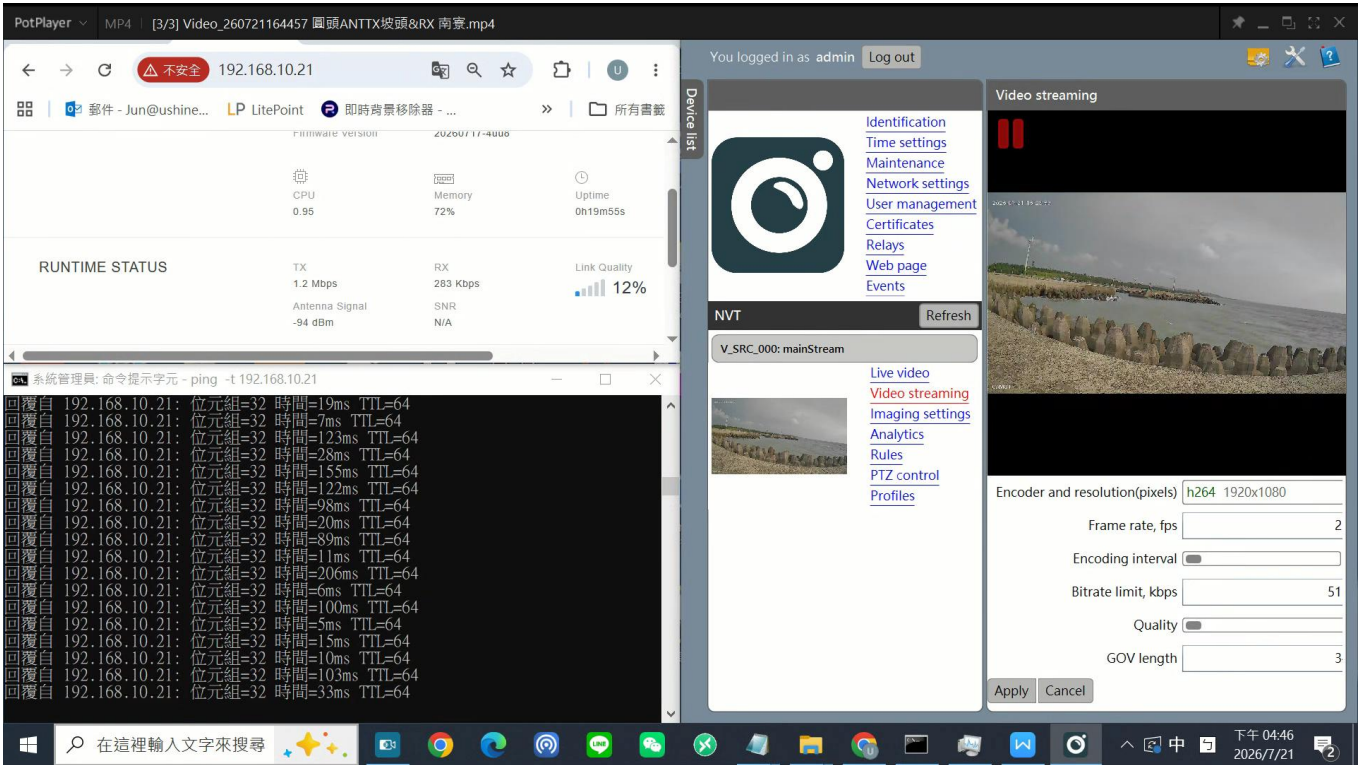


Figure 7-1. Runtime screenshot A supplied with the test package

Visible indicator	Captured value
Visible video reference	The media-player title includes the 9.78 km omni-antenna test video filename.
Displayed device address	192.168.10.21
Runtime TX / RX	1.2 Mbps / 283 Kbps
Antenna signal	-94 dBm
Link quality	12%
SNR	N/A
Visible ping samples	Approximately 5 ms to 206 ms in the captured window; no timeout is visible in the screenshot.

The values above are instantaneous UI readings, not a continuous measurement log. TX/RX counters are presented from the perspective of the displayed device and should not be treated as an application-layer throughput benchmark without confirming device role and counter definition.

07A

RUNTIME EVIDENCE - SECOND CAPTURE

Higher signal and SNR state shown in the supplied evidence

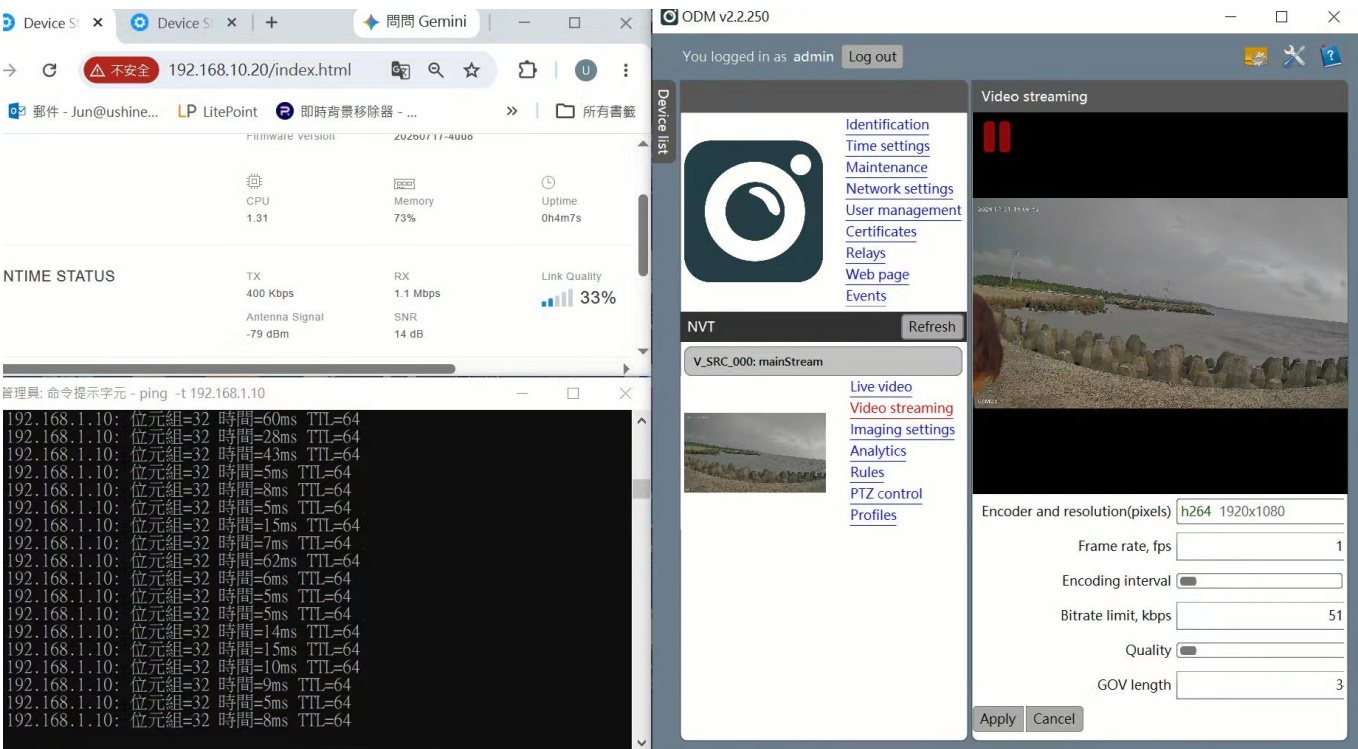


Figure 7-2. Runtime screenshot B supplied with the test package

Visible indicator	Captured value
Displayed device address	192.168.10.20
Runtime TX / RX	400 Kbps / 1.1 Mbps
Antenna signal	-79 dBm
Link quality	33%
SNR	14 dB
Visible ping samples	Approximately 5 ms to 60 ms in the captured window; no timeout is visible in the screenshot.
Video interface	H.264, 1920 × 1080 shown in the ODM interface

Compared with Figure 7-1, this capture shows a materially stronger signal and lower visible latency variation. The source package does not provide synchronized timestamps or a continuous log, so the screenshots should be treated as representative states rather than a controlled statistical comparison.

08

ENGINEERING INTERPRETATION

Clearly separated from the source-derived observations

Interpretation of the 9.78 km results

- The directional antenna result indicates that additional antenna gain and focused energy materially improve the link margin at the longest tested distance.
- The omni antenna remaining smooth at 512 kbps but showing minor stuttering at 1024 kbps is consistent with a reduced fade margin: a higher application bitrate leaves less tolerance for retransmissions and latency variation.
- The -94 dBm / 12% screenshot represents a weak-link state. The visible ping variation up to approximately 206 ms is consistent with retransmissions, queueing or temporary channel degradation, but the screenshot alone cannot identify the exact cause.
- The -79 dBm / 14 dB SNR screenshot represents a stronger state and shows a narrower visible ping range. This supports the practical value of better alignment, antenna gain and path clearance.

Customer deployment guidance

Use case	Recommendation
For maximum stability at ~10 km	Use the directional antenna, verify alignment at both ends and keep the video bitrate at or below the validated 1024 kbps setting.
When omni coverage is required	Use 512 kbps as the source-supported stable setting at 9.78 km; validate 1024 kbps at the final installation before release.
Installation	Increase antenna height where practical, minimize cable loss and maintain clear polarization/orientation.
Future validation	Record continuous RSSI/SNR, packet loss, ping statistics, UDP/TCP throughput, weather, GPS coordinates, antenna gain and spectrum occupancy.

These recommendations are engineering inferences based on the supplied observations and should be revalidated for different hardware revisions, antennas, regulatory power limits, terrain and interference conditions.

09

CONCLUSION AND
RECOMMENDATIONS

Customer-facing outcome statement

3.86 KM Smooth Directional AP / omni STA	9.78 KM DIRECTIONAL Smooth 512 and 1024 kbps	9.78 KM OMNI Smooth at 512 Minor stutter at 1024
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Conclusion: The supplied evidence demonstrates practical long-range operation of the 5.8 GHz video transmission system at 3.86 km and 9.78 km under the documented field configurations. The directional antenna provided the strongest long-range result, maintaining smooth video at both 512 kbps and 1024 kbps over 9.78 km. The omni configuration remained smooth at 512 kbps and showed minor stuttering at 1024 kbps over the same route.

Recommended customer configuration: For deployments approaching 10 km, use the directional antenna and a validated video bitrate of 512-1024 kbps. Where omni coverage is required, 512 kbps is the source-supported stable setting at the tested distance.

Qualification statement: These results apply to the tested firmware, radio settings, antenna arrangements and field conditions. They are not a guarantee of identical performance in other interference, terrain, weather or regulatory environments.

Recommended next test package

- 10 km repeat test with synchronized continuous RSSI, SNR, ping and packet-loss logging.
- UDP throughput and application video bitrate captured simultaneously.
- Directional antenna gain, cable loss and verified EIRP documented.
- Photographs of antenna orientation and measured installation heights at both endpoints.
- Weather, wind, spectrum scan and exact coordinates recorded in the test sheet.

A

APPENDIX A - REFERENCED MEDIA

Files named in the original source report

Media item	Reference
3.86 km video	Video_260721155939T TX 天德堂&RX 水月橋上.mp4
9.78 km directional video	Video_260721162958 指向天線 TX 坡頭&RX 南寮.mp4
9.78 km omni video	Video_260721164457 圓頭 ANTTX 坡頭&RX 南寮.mp4
Runtime screenshot A	Media-player title references the 9.78 km omni-antenna video.
Runtime screenshot B	Device and ONVIF runtime status capture.

The three video files are referenced by name in the source report but were not embedded in the supplied PDF. This Word document includes the two runtime screenshots provided in the conversation.



Shuiyue Bridge map detail



Tiande Temple map detail



Nanliao Fishing Port map detail



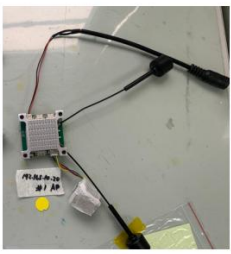
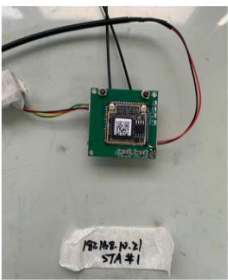
Potou Fishing Port map detail

B1

APPENDIX B - ORIGINAL SOURCE PAGE

Original field test record, page 1 of 3

宋總 5.8G 圖傳遠距離測試
TEST DATE: 202607021@TX 天德堂, RX 水月橋&TX 坡頭漁港 ,RX 南賽漁港
TEST SAMPLE:
FW:fw-20260717-4uu8.bin

AP 端 (ANT:圓頭&指向天線)		
STA 端 + CAMERA (ANT:圓頭)	 	
EUT setup	Frequency:36 Channel:5.8G Baud Rate:115200 Bandwidth:20M MCS:MCS0 Power:31.5	

B2

APPENDIX B - ORIGINAL SOURCE PAGE

Original field test record, page 2 of 3

測試地點:

AP@水月橋上	STA@天德堂	3.86km
		
AP@南寮漁港	STA@坡頭漁港	9.78km
		

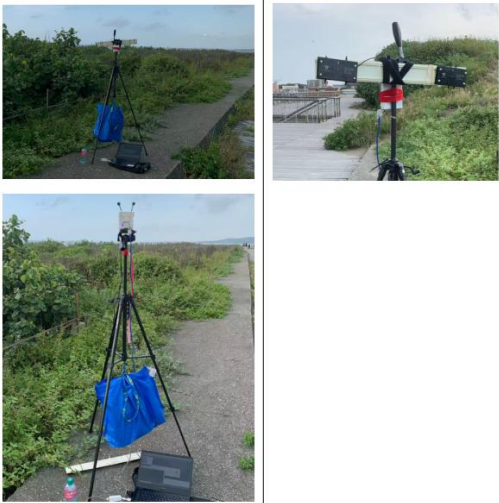
TEST LOCALTION #1 : AP@水月橋上 , STA@天德堂

AP 端 @水月橋上 架高約 1.5m 海拔約 8M 指向天線		
STA 端 @天德堂 架高約 1.5m 海拔約 85M 圓頭 ANT		
Result	影像流暢  Video_26072115 5939T TX天德堂& 請參考 VIDEO	

B3

APPENDIX B - ORIGINAL SOURCE PAGE

Original field test record, page 3 of 3

TEST LOCALTION #2 : AP@南寮漁港,STA@坡頭漁港		
AP 端 @南寮漁港 架高約 1.5m (ANT:圓頭&指向天線)		
STA 端 @坡頭漁港 架高約 1.5m		
Result	指向天線:Bitrate : 512 &1024, 影像流暢  Video_26072116 2958 指向天線 TX 圓頭 ANT:Bitrate : 512 影像流暢 , 1024 些微卡頓  Video_26072116 4457 圓頭ANTTX	